

A Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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Abstract

We determine the axial Schwarzschild spin-two extension and one of its quasinormal resonances in four-dimensional pure Weyl gravity. The reduced six-state Bach system has an exact filtration with two spin-two Regge–Wheeler factors and one spin-one Maxwell factor. The repeated spin-two block is not a rational direct sum: it is a non-split self-extension and the first parameter jet of a lower-dimensional family. After Liouville reduction, its projective class is represented by

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[1 - \frac{2}{r} \right].$$

The class in brackets is exactly the transverse-traceless critical Einstein–Weyl mass direction. The required rational gauge has the linearly growing asymptotics of the differentiated massive Jost phase; the long-range Coulomb contribution cancels the apparent logarithmic phase derivative. Thus the local Bach jet and the differentiated physical Jost germs agree, up to endpoint normalization.

At a validated simple Schwarzschild quasinormal frequency, the connection shear is nonzero and the spin-one factor is a local unit. The complete connection therefore has Smith type $(0, 0, 2)$. Its geometric root is Einstein, while the generalized root has a nonzero gauge-invariant Ricci-carrier component. The same invariant is the nonzero velocity of the massive spin-two quasinormal branch. We prove that the compact-source, compact-observation outgoing radial Green operator on the Schwarzschild exterior has a nonzero rank-one second-order pole.

Exact axial reconstruction shows that the Einstein range of the leading pole has nonzero Bondi shear, and an explicit conserved, traceless, compactly supported odd source has nonzero adjoint overlap. Hence both source and observation channels can be nonvanishing. The isolated resonance contour contains the expected polynomially weighted Einstein profile. We do not claim a global retarded quasinormal expansion: the time-independent generalized component has weakened $O(1)$ strain, and a closed weighted Fredholm realization with causal contour control remains open.

1 Introduction

Fourth-order linear equations are often described by listing their scalar factors. That description is incomplete when a factor is repeated. The layers may form a direct sum, or they may form a non-semisimple extension. Only the second possibility carries an intrinsic generalized state and can turn a simple resonance of the repeated factor into a defective resonance of the full system.

This distinction is realized by axial Schwarzschild perturbations of pure Weyl gravity. In the factor-adapted radial variables, the Bach equation contains two spin-two Regge–Wheeler layers and one spin-one Maxwell layer, but the two spin-two layers cannot be separated by a rational local field redefinition. The extension is also the critical limit of the massless and massive spin-two branches

of Einstein–Weyl gravity. At a Schwarzschild quasinormal frequency, this confluence produces a length-two root chain rather than two independent Einstein roots.

The paper has five results.

1. The axial spin-two Bach block is a non-split rational Regge–Wheeler self-extension and an exact first parameter jet.
2. Its projective cocycle has the elementary mass-direction normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[1 - \frac{2}{r} \right],$$

and the equivalence extends to differentiated horizon and infinity Jost germs.

3. A validated QNM calculation selects the defective Smith type $(0, 0, 2)$. The generalized root necessarily crosses the Einstein filtration and is the normalization-independent tangent class of the massive QNM.
4. The compact-source, compact-observation exterior outgoing radial Green operator has a genuine rank-one pole of order two.
5. Exact metric reconstruction and an explicit conserved, traceless odd source show that the range and source covectors of this pole can both be nonzero.

These statements combine exact algebra with validated numerics. The module reduction, cocycle identities, mass comparison, Green reduction, metric reconstruction, and source construction are exact. The existence and uniqueness of the enclosed QNM, nonvanishing of the projective selector, and spin-one local-unit condition are interval-validated.

Claim boundary

Result	Input	Status
Filtration, partial jet, and nonsplitting	Rational identities and exhaustive coboundary obstruction	LOCAL-ALGEBRAIC
Mass-direction and Jost-germ comparison	Exact cocycle primitive and moving massive phases	LOCAL-ALGEBRAIC
Defective QNM and non-Einstein root	Validated Evans winding, selector, spin-one unit	REDUCED-MODE
Exterior cut-off Green double pole	Analytic Jost maps and exact transparent cuts	REDUCED-MODE
Bondi-shear and compact-source nonannihilation	Exact reconstruction and conserved-source identities	REDUCED-MODE
Global retarded ringdown expansion	Weighted Fredholm domains and contour deformation	Not established

The last row is essential. The paper proves a meromorphic exterior radial response after compact source and observation localization, and it proves the polynomial contribution of an isolated local resonance contour. It does not prove a complete causal quasinormal expansion, late-time asymptotics, detector sensitivity, or excitation by a specified astrophysical matter model.

2 The filtered axial Bach module

2.1 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \quad D = f\partial_r, \quad f = 1 - \frac{2}{r},$$

with $\omega \neq 0$ unless stated otherwise. The spin-two and spin-one scalar operators for axial $\ell = 2$ are

$$L_2 = D^2 + U_2, \quad U_2 = \omega^2 - f \left(\frac{6}{r^2} - \frac{6}{r^3} \right),$$

$$L_1 = D^2 + U_1, \quad U_1 = \omega^2 - f \frac{6}{r^2}.$$

We use $D(q)$ and $D(U)$, rather than primes, for the radial derivation. A prime on a connection coefficient below denotes ∂_ω .

The Einstein submodule is the kernel of the gauge-invariant traceless-Ricci, equivalently Schouten, carrier. Thus “non-Einstein” means nonzero projection to this carrier quotient, not failure of a coordinate gauge choice.

2.2 Filtration and partial jet

In the certified factor gauge, the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1Z, \end{aligned} \tag{1}$$

where $X, Y, Z \in K^2$, while A and A_1 are the companion matrices of L_2 and L_1 . The rational reconstruction into the axial metric and curvature variables has full row rank on the declared domain.

Theorem 2.1 (Filtered axial Bach module). *The reduced axial $\ell = 2$ Bach module has a filtration with successive quotients*

$$M_2, \quad M_2, \quad M_1.$$

The middle four-state object is an extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0,$$

and the final quotient is the spin-one Maxwell module M_1 .

This theorem is an exact rational statement. It does not rely on an endpoint normalization or on the QNM calculation.

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}, \quad \Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

Theorem 2.2 (Exact partial jet). *The system (1) is the first spin-two-row jet of $\mathcal{B}(\tau)$ at $\tau = 0$. Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \quad \det \Phi_6 = (\det P)^2 \det R.$$

Every factor-adapted endpoint connection consequently has the form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \quad \det T = a^2 g. \quad (2)$$

Proof. Differentiate $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$ at $\tau = 0$. The differentiated first row, the undifferentiated spin-two row, and the spin-one row reproduce (1). The determinant and connection identities follow from block triangularity. $\square$

2.3 Projective extension class

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \quad Ly = 0, \quad L = D^2 + U.$$

Define

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \quad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Lemma 2.3 (Triangular gauge). *For $q \in K$, set*

$$Q(q) = qD - \frac{1}{2}D(q).$$

On $\ker L$,

$$[L, Q(q)] = -\frac{1}{2}\mathcal{K}_U(q).$$

Thus the lift change $x \mapsto x + Q(q)y$ changes the scalar density by

$$\mathcal{I} \mapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

Proof. Expand the commutator by the Leibniz rule and use $D^2y = -Uy$. The coefficient of Dy cancels; the remaining coefficient is $-[D^3q + 4UDq + 2D(U)q]/2$. $\square$

The projective extension class is therefore

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

The operator $\mathcal{K}_U$ is the symmetric-square operator of L .

Theorem 2.4 (Mass-direction normal form). *For the axial $\ell = 2$ Bach extension,*

$$\mathcal{I}_{\text{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

Let

$$q(r, \omega) = -\frac{i}{120\omega} \left(15r + 13 + \frac{12}{r} + \frac{9}{r^2} \right).$$

Then, for $\omega \neq 0$,

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2} q + \frac{i\omega}{2} f, \quad (3)$$

and hence

$$\boxed{[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} [f].}$$

Proof. Substitute $D = f\partial_r$, U_2 , and the displayed q into $\mathcal{K}_{U_2} q$. After clearing $120r^4\omega$, the assertion is a polynomial identity in r and ω . Exact coefficient comparison gives (3). $\square$

Theorem 2.5 (Non-split physical-axis extension). *For every real $\omega > 0$, the extension E_2 is non-split over $\mathbb{C}(r)[D]$.*

Certificate-supported exact proof. The rational splitting equation has exhaustive pole and degree bounds and reduces to a finite coefficient system. Fixed minors of its coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456\omega^{10}, \quad \det[M \mid \text{rhs}]_{[0,1,2,5]} = -645120i\omega^9.$$

They are simultaneously nonzero for every real $\omega > 0$, so the inhomogeneous coboundary equation is inconsistent. $\square$

Theorem 2.5 is a statement about the differential module; Theorem 2.4 identifies a simple representative of its class. Neither theorem asserts that every QNM of the repeated scalar factor must be defective. Defectiveness is a resonant evaluation of the nonzero extension class.

3 Critical-mass interpretation

3.1 The transverse-traceless mass jet

On the transverse-traceless spin-two sector, introduce the Einstein–Weyl family

$$(\mathcal{E} + m\mathcal{A})\phi = 0.$$

Its axial scalar equation is

$$L_m(\omega)y = [D^2 + \omega^2 - V_2 - mf]y = 0. \quad (4)$$

For a differentiable solution family,

$$L_2 \partial_m y = f y \quad (m = 0).$$

Theorem 3.1 (Exact critical-mass jet). *The projective class of the transverse-traceless mass jet is*

$$[\mathcal{I}_{\text{mass}}] = [f].$$

Consequently,

$$\boxed{[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} [\mathcal{I}_{\text{mass}}], \quad m = \frac{i\omega}{2} \tau.}$$

The parameter identity is understood to first order and modulo rational triangular gauge.

Proof. Differentiating (4) gives the extension density f . The result follows from Theorem 2.4. $\square$

3.2 Endpoint-compatible gauge

Because Lemma 2.3 contains a factor $1/2$, the field redefinition that removes $\mathcal{K}_{U_2}q$ is

$$Q_q = 2qD - D(q).$$

On $\ker L_2$,

$$[L_2, Q_q] = -\mathcal{K}_{U_2}(q).$$

Thus, if $L_2x = \mathcal{I}_{\text{Bach}}y$, then

$$\widehat{x} = x + Q_qy \implies L_2\widehat{x} = \frac{i\omega}{2}fy.$$

The primitive has the forced infinity behavior

$$q(r, \omega) = -\frac{i}{8\omega}r + O(1).$$

This growth is not an endpoint obstruction. It is the derivative of the moving massive phase.

Theorem 3.2 (Differentiated Jost compatibility). *Let $y_\sigma(m, r)$, $\sigma = \pm 1$, be a massive Jost germ at infinity with*

$$k = (\omega^2 - m)^{1/2}, \quad y_\sigma(m, r) = e^{\sigma i k r} r^{\rho_\sigma(m)} (1 + O(r^{-1})),$$

$$\rho_\sigma(m) = \sigma i \left(2k + \frac{m}{k} \right).$$

Then

$$\rho'_\sigma(0) = 0, \quad \frac{\partial_m y_\sigma(0, r)}{y_\sigma(0, r)} = -\frac{\sigma i}{2\omega}r + O(1).$$

With $c(\omega) = i\omega/2$,

$$\frac{Q_q y_\sigma}{y_\sigma} = \frac{\sigma r}{4} + O(1) = c(\omega) \frac{\partial_m y_\sigma(0, r)}{y_\sigma(0, r)} + O(1).$$

At the horizon, the mass perturbation is multiplied by $f = O(r - 2)$, so the indicial exponent is unchanged and Q_q preserves the analytic ingoing germ. At each endpoint e ,

$$\widehat{x}_e - c(\omega) \partial_m y_e(0) = \lambda_e(\omega) y_e$$

for an analytic scalar normalization λ_e .

Proof. The infinity equation is

$$D^2y + \left(k^2 + \frac{2m}{r} + O(r^{-2}) \right) y = 0.$$

The coefficient of r^{-1} gives

$$2\sigma i k \rho_\sigma + 4k^2 + 2m = 0.$$

Since $k'(0) = -1/(2\omega)$,

$$\frac{d}{dm} \left(2k + \frac{m}{k} \right) \Big|_{m=0} = -\frac{1}{\omega} + \frac{1}{\omega} = 0.$$

The Coulomb-power derivative therefore cancels, while differentiation of the plane-wave phase gives the linear term. Finally, $Dy_\sigma/y_\sigma = \sigma i\omega + O(r^{-1})$ and $q = -ir/(8\omega) + O(1)$ give the stated asymptotics of $Q_q y_\sigma$. The difference between $\widehat{x}$ and $c \partial_m y$ is homogeneous and belongs to the same one-dimensional endpoint Jost class. At the horizon the claim follows from regular Frobenius dependence and the unchanged indicial exponent. $\square$

Corollary 3.3 (Evans derivative and mass velocity). *Let $a(\omega, m)$ be the massive spin-two Evans coefficient in analytic Jost normalizations, and let $b_B(\omega)$ be the Bach connection shear. There is an analytic $h(\omega)$ such that*

$$b_B(\omega) = \frac{i\omega}{2} \partial_m a(\omega, 0) + h(\omega) a(\omega, 0).$$

At a simple QNM ω_n ,

$$b_B(\omega_n) = \frac{i\omega_n}{2} \partial_m a(\omega_n, 0).$$

Writing

$$\kappa_n = \frac{b_B(\omega_n)}{a_\omega(\omega_n, 0)},$$

the massive QNM branch satisfies

$$\boxed{\omega'_n(0) = \frac{2i}{\omega_n} \kappa_n.} \tag{5}$$

Proof. The endpoint corrections in Theorem 3.2 are scalar changes of differentiated Jost normalizations, hence contribute an analytic multiple of a . At a zero of a , this term vanishes. Differentiate $a(\omega_n(m), m) = 0$ and solve for $\omega'_n(0)$. $\square$

This is the asymptotically flat QNM analogue of the mass-derivative construction of logarithmic partners in critical gravity [1, 2]. The terminology must be used with care: the Schwarzschild endpoint tangent is polynomial, not an independent radial $\log r$ mode.

4 The defective Schwarzschild resonance

4.1 Smith dichotomy

At a simple spin-two zero, analytic elimination of the spin-one unit reduces (2) to

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \quad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

Proposition 4.1 (Local Smith dichotomy). *If $g(\omega_n) \neq 0$, then*

$$\text{Smith}(T) = \begin{cases} (0, 0, 2), & b(\omega_n) \neq 0, \\ (0, 1, 1), & b(\omega_n) = 0. \end{cases}$$

The first case has one length-two root chain; the second has two independent length-one roots.

Proof. With $z = \omega - \omega_n$, $a = a_1 z + O(z^2)$, $a_1 \neq 0$. The determinant valuation of the spin-two block is two. If $b(\omega_n) \neq 0$, its first Fitting ideal is the unit ideal and the exponents are $(0, 2)$. If $b(\omega_n) = 0$, all entries vanish to first order and the exponents are $(1, 1)$. The spin-one unit contributes a zero exponent. $\square$

4.2 Validated specialization

The positively oriented validated contour encloses

$$\Omega_n = \{\omega : |\omega - (-0.3736716844180418 + 0.0889623156889357i)| \leq 10^{-7}\}.$$

Projective interval panels exclude zeros on the contour and give winding number +1. The disk therefore contains exactly one simple spin-two zero. On the localized disk,

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation bounds the spin-one mismatch modulus below by 1/2500.

Theorem 4.2 (Certified defective resonance). *At the unique QNM in Ω_n , the complete axial connection has Smith valuations*

$$(0, 0, 2).$$

Its spin-two divisor has algebraic multiplicity two, geometric multiplicity one, and one length-two root chain.

Proof. The winding certificate supplies the simple spin-two zero. The selector enclosure gives $b(\omega_n) \neq 0$, and the spin-one bound gives $g(\omega_n) \neq 0$. Proposition 4.1 applies. $\square$

4.3 One resonant invariant

Encode the endpoint conditions in an analytic finite-interval pencil

$$L(\omega) : X \longrightarrow Y.$$

Choose right and left states at the simple scalar resonance,

$$L_n u_n = 0, \quad L_n^* \tilde{u}_n = 0,$$

and define, with all augmented endpoint contributions included,

$$\alpha_n = \langle \tilde{u}_n, L_n' u_n \rangle \neq 0.$$

For the repeated block

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

set

$$\beta_n = \langle \tilde{u}_n, K_n u_n \rangle.$$

Theorem 4.3 (Resonant evaluation). *In endpoint frames compatible with the partial jet,*

$$\kappa_n = \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = - \left. \frac{d\omega_n}{d\tau} \right|_0 = - \frac{i\omega_n}{2} \left. \frac{d\omega_n}{dm} \right|_0. \quad (6)$$

The ratio is invariant under a common analytic nonzero Evans unit and under triangular changes of splitting.

Proof. Differentiate

$$[L(\omega) + \tau K(\omega)]u(\tau) = 0$$

along the resonance branch. Pairing with $\tilde{u}_n$ gives

$$\omega'_n(0)\alpha_n + \beta_n = 0.$$

Differentiating the connection divisor gives

$$a'(\omega_n)\omega'_n(0) + b(\omega_n) = 0,$$

because the partial jet identifies $b = \partial_\tau a$. The mass identity follows from (5). A triangular change $K \mapsto K + LQ - QL$ has zero resonant pairing, and an Evans unit multiplies numerator and denominator by the same nonzero value. $\square$

Proposition 4.4 (Non-Einstein generalized root). *The geometric root belongs to the Einstein submodule. Every length-two root polynomial has generalized coefficient with nonzero projection to the Ricci-carrier quotient. In the normalized triangular frame, that projection is*

$$\boxed{-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n} \neq 0.}$$

Proof. Write $a = a_1 z + O(z^2)$, $b = b_0 + O(z)$, with $a_1 b_0 \neq 0$. For a root polynomial

$$c(z) = c_0 + z c_1 + O(z^2), \quad T_2(z)c(z) = O(z^2),$$

the order-zero equation gives $c_0 = e_X$, the Einstein vector. Writing $c_1 = x e_X + y e_Y$, the order- z equation is

$$a_1 + b_0 y = 0.$$

Hence $y = -a_1/b_0 \neq 0$. Adding an Einstein vector changes x , not the carrier coefficient y . $\square$

Thus the defective resonance is not the same Einstein QNM counted twice. Its generalized member is the canonical mass-tangent class modulo the ordinary QNM.

5 Exterior Green pole and critical response

5.1 Finite cuts and the exterior inverse

Put $x = r_*$. On the certified QNM neighborhood, let

$$u_H(x, \omega), \quad u_\infty(x, \omega)$$

be analytic spin-two Jost solutions satisfying the future-horizon and outgoing-infinity conditions. Their Wronskian is the scalar Evans coefficient, up to an analytic unit.

Choose nontrivial smooth source and observation cutoffs χ_s, χ_o and finite cuts

$$x_- < \text{supp } \chi_s \cup \text{supp } \chi_o < x_+.$$

Impose the exact transparent boundary conditions

$$W(u, u_H)|_{x_-} = 0, \quad W(u, u_\infty)|_{x_+} = 0.$$

They are analytic in ω , and uniqueness of the exterior Jost problems identifies the finite-interval inverse between the cutoffs with $\chi_o R_{\text{ext}}(\omega) \chi_s$.

Theorem 5.1 (Exterior cut-off Green pole). *For any such nontrivial compact source and observation cutoffs,*

$$\chi_o R_{\text{ext}}(\omega) \chi_s$$

has a nonzero rank-one pole of order two at ω_n . In the augmented normalization,

$$\boxed{\text{Coeff}_{(\omega-\omega_n)^{-2}}(\chi_o R_{\text{ext}} \chi_s) = -\frac{\beta_n}{\alpha_n^2} (\chi_o u_n) \otimes (\tilde{u}_n \chi_s).} \quad (7)$$

Proof. The repeated block inverse is

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1} K L^{-1} \\ 0 & L^{-1} \end{pmatrix}.$$

At a simple scalar resonance,

$$L(\omega)^{-1} = \frac{u_n \otimes \tilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block therefore has principal coefficient

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \tilde{u}_n.$$

The exact transparent conditions identify this finite-interval coefficient with the cut-off exterior coefficient. It is nonzero because $\beta_n \neq 0$, and a nonzero solution of a second-order ODE cannot vanish on a nonempty open interval. $\square$

The full doubled-state principal coefficient maps the carrier channel to the Einstein channel and vanishes on the Einstein channel; it is square-zero on that graded space. Its projection to a single metric or radial block is rank one but is not intrinsically a nilpotent operator after the grading has been forgotten.

5.2 Mass derivative of the parent response

At points where the inverses exist,

$$\partial_m(\mathcal{E} + m\mathcal{A})^{-1} = -(\mathcal{E} + m\mathcal{A})^{-1} \mathcal{A}(\mathcal{E} + m\mathcal{A})^{-1}.$$

The metric–metric block of the pure-Weyl parent inverse is therefore

$$\boxed{G_{hh}^W(\omega) = -\frac{1}{4\alpha_W} \partial_m(\mathcal{E} + m\mathcal{A})^{-1} \Big|_{m=0}.} \quad (8)$$

No commutativity assumption is used.

If the massive second-order inverse has the local simple-pole form

$$R_m(\omega) = \frac{P_n(m)}{\omega - \omega_n(m)} + H_n(\omega, m),$$

then (8) gives

$$G_{hh}^W(\omega) = -\frac{1}{4\alpha_W} \left[\frac{\omega'_n(0) P_n(0)}{(\omega - \omega_n)^2} \frac{P'_n(0)}{\omega - \omega_n} \partial_m H_n(\omega, 0) \right].$$

Thus the double pole measures QNM motion; the accompanying simple pole measures motion of the residue projector.

Corollary 5.2 (Isolated resonance contour). *For a contour enclosing only the certified local resonance,*

$$\frac{1}{2\pi i} \oint e^{i\omega t} G_{hh}^W(\omega) d\omega = -\frac{e^{i\omega_n t}}{4\alpha_W} [P'_n(0) + it\omega'_n(0)P_n(0)].$$

The t -weighted spatial profile is the ordinary Einstein QNM. The constant generalized component contains the non-Einstein mass tangent.

This is a local residue theorem, not a global retarded expansion. The latter would require a Laplace contour, a closed weighted realization, control of thresholds and non-pole contributions, and a justified contour deformation.

6 Physical source and observation channels

6.1 Metric reconstruction at null infinity

Let E be the unit outgoing Einstein line and R the unit outgoing carrier line in the exact factor frame. After factoring the common retarded phase $e^{i\omega u}$, their Regge–Wheeler-gauge metric heads are

$$(H_0, H_1)_E = (-r, 2r) + O(1),$$

$$(H_0, H_1)_R = \left(\frac{3}{4}r^2 - \frac{3}{2}r, -\frac{3}{2}r^2\right) + O(1).$$

Hence

$$(H_0, H_1)_R = -\frac{3}{4}r(H_0, H_1)_E + O(E).$$

In odd radiation gauge,

$$h_u \mapsto h_u - i\omega\xi, \quad h_2 \mapsto h_2 - 2\xi, \quad \xi = \frac{h_u}{i\omega},$$

so $h_2 = 2ih_u/\omega$. Writing $h_{AB} = h_2 X_{AB}$, define the Bondi-shear observation map by

$$\mathcal{O}_{\mathcal{J}^+} h = \lim_{r \rightarrow \infty} \frac{h_{AB}}{r}.$$

Theorem 6.1 (Null-infinity reconstruction). *The outgoing Einstein line has nonzero Bondi shear,*

$$\boxed{\mathcal{O}_{\mathcal{J}^+} E = -\frac{2i}{\omega} X_{AB} \neq 0.} \tag{9}$$

Its strain has the standard radiative falloff

$$\frac{h_{AB}^E}{r^2} = -\frac{2i}{\omega r} e^{i\omega u} X_{AB} + O(r^{-2}).$$

The carrier line instead has

$$\boxed{\frac{h_{AB}^R}{r^2} = \frac{3i}{2\omega} e^{i\omega u} X_{AB} + O(r^{-1}).} \tag{10}$$

Therefore the time-independent generalized component has weakened $O(1)$ strain in this standard odd radiation gauge. The Einstein range of the double-pole coefficient is nevertheless observed nontrivially at future null infinity.

Proof. Substitute the exact outgoing factor-frame series into the algebraic metric reconstruction. The odd gauge law then gives (9) and (10). Adding an Einstein representative cannot cancel the carrier r^2 coefficient, which is one power higher. The trace-free odd tensor harmonic cannot be removed by a Weyl rescaling; its removal would require a large boundary diffeomorphism rather than a standard small gauge transformation. $\square$

Combining Theorems 5.1 and 6.1, the principal Bondi-shear coefficient is nonzero. In the parent normalization it is

$$\mathcal{O}_{\mathcal{J}^+} G_{-2} = \frac{i\omega'_n(0)}{2\alpha_W \alpha_n \omega_n} X_{AB} \otimes \tilde{u}_n.$$

The isolated contour therefore contains an ordinary u/r radiative profile multiplied by the adjoint source overlap.

6.2 A conserved, traceless source with nonzero overlap

Let $\ell \geq 2$, $\mu = (\ell - 1)(\ell + 2)$, and $\omega \neq 0$. Use the Martel–Poisson odd source projections and orientation $\epsilon_{tr} = 1$, $\epsilon^{tr} = -1$ [3].

Theorem 6.2 (Conserved source nonannihilation). *For any*

$$F \in C_c^\infty((2M, \infty)),$$

define

$$P_t = 0, \quad P_r = \frac{\mu F}{2i\omega r f}, \quad P = \frac{\partial_r(rF)}{2i\omega}.$$

Then

$$\nabla_a P^a + \frac{2}{r} r_a P^a - \frac{\mu}{r^2} P = 0,$$

and the Cunningham–Price–Moncrief source is

$$S_{\text{odd}} = -\frac{2r}{\mu} \epsilon^{ab} \nabla_a P_b = \frac{F}{f}.$$

The associated odd stress tensor

$$T^{aB} = \frac{P^a}{16\pi r^2} X^B, \quad T^{AB} = \frac{P}{8\pi r^4} X^{AB},$$

with all other harmonic components zero, is smooth and radially compact, and satisfies

$$\boxed{\nabla_\mu T^{\mu\nu} = 0, \quad T^\mu{}_\mu = 0.}$$

The complexified conserved-traceless odd source map is therefore onto the smooth compact reduced radial forcings.

Moreover, choose a nonnegative nonzero $\eta \in C_c^\infty((2M, \infty))$ supported where the adjoint state $\tilde{u}_n$ is nonzero and set

$$F = \eta \overline{\tilde{u}_n}.$$

Then the augmented pairing has no endpoint term and

$$\boxed{\langle \tilde{u}_n, F \rangle_{\text{aug}} = \int \eta |\tilde{u}_n|^2 dr_* > 0.}$$

Thus a smooth compact conserved and traceless odd source excites the certified double pole.

Proof. Since $\sqrt{-\det g_{ab}} = 1$,

$$P^r = f P_r = \frac{\mu F}{2i\omega r}, \quad \nabla_a P^a = \partial_r P^r.$$

Direct substitution proves the reduced conservation identity and $f S_{\text{odd}} = F$. The remaining spacetime conservation equations vanish by the divergence-free odd vector harmonic, and tracelessness follows from $T^{ab} = 0$ and $\Omega_{AB} X^{AB} = 0$. Compact support removes all endpoint terms from the adjoint pairing, and the final integral is strictly positive. $\square$

Remark 6.3 (What the source theorem does not say). The theorem constructs a conformally admissible external source and proves nonannihilation. It does not identify a specified material trajectory or an energy-condition-satisfying matter model. In particular, a standard massive point-particle stress tensor is not trace-free and cannot be coupled directly to the Bach equation without a conformal completion or a compensator.

7 Scope, evidence, and open problem

7.1 The remaining analytic problem

The exterior cut-off theorem is already a differential-response result on the Schwarzschild exterior. The remaining causal problem is narrower but substantive. A global realization must provide:

1. closed weighted domains implementing horizon-ingoing and differentiated outgoing Jost conditions;
2. a Fredholm pencil of index zero with a fixed analytic domain;
3. bounded reconstruction through the physical gauge quotient;
4. a causal Laplace contour and a justified deformation through the QNM disk;
5. control of threshold, branch-cut, and non-pole contributions.

The weakened falloff in (10) explains why this is not routine. A standard outgoing asymptotically flat function space may contain the Einstein root but exclude its mass tangent. One may instead use an enlarged tangent domain or keep the bulk space fixed and encode the differentiated endpoint germ as a finite-dimensional boundary component. This paper uses the latter construction locally but does not complete its global causal implementation.

7.2 Certificate dependency

Certificate family	Imported statement	Type
Axial triangular preflight	Filtration, factor gauge, and reconstruction rank	Exact
Projective cocycle	Cocycle, primitive, and rational nonsplitting obstruction	Exact
Critical mass jet	Scalar mass direction and parent inverse identity	Exact
Analytic continuation	Analytic horizon and infinity Jost germs on the QNM disk	Exact gate
Full Evans contour	Zero-free contour and winding +1	Validated
Local selector	Nonzero projective tangent on the root disk	Validated
Spin-one local unit	Nonzero spin-one mismatch	Validated
Fredholm promotion	Finite-interval Poisson/trace reduction	Exact reduction
Null-infinity reconstruction	Metric heads, radiation gauge, and Bondi shear	Exact
Conserved source	Conservation, tracelessness, and adjoint overlap	Exact

The generated claim map pins the authority files and their hashes. Its verifier independently checks the displayed cocycle identity, the mass-parameter normalization, the Smith and Green formulas, the reconstruction falloffs, the source conservation law, and explicit non-results. Missing data, a stale hash, an unsupported domain, or a failed mutation is a rejection rather than a partial pass.

Explicit non-results. This paper does not establish a global causal spacetime resolvent, a complete QNM expansion, a late-time asymptotic formula, a retarded inverse-Laplace deformation, a time-domain instability, finite Bondi flux for the generalized constant component, a standard asymptotically flat domain containing that component, excitation by a specified astrophysical source, detector sensitivity, an overtone tower, an all-multipole Bach nonsplitting theorem, or a quantum positivity/no-go theorem. The polynomially weighted term is proved only for the isolated local resonance contour.

8 Conclusion

The repeated Regge–Wheeler factor in axial Schwarzschild pure Weyl gravity is not a direct sum. It is a non-split self-extension with the elementary projective class

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[1 - 2/r].$$

That direction is exactly the critical transverse-traceless mass jet. The linearly growing rational equivalence gauge is compatible with the differentiated physical Jost germs because the massive Coulomb exponent has vanishing first derivative.

At the certified damped Schwarzschild mode, the scalar QNM is simple but the extension selector is nonzero. The full connection is therefore defective. Its geometric root is Einstein; its generalized root has a nonzero Ricci-carrier quotient and is the massive-QNM tangent modulo the ordinary root. The same invariant controls the nonzero rank-one second-order pole of the exterior cut-off outgoing Green operator.

The leading pole survives the two physically relevant channel tests. Its Einstein range has nonzero Bondi shear, and a smooth compact conserved, traceless odd source can have strictly nonzero adjoint overlap. The enhanced isolated-contour profile is therefore an ordinary outgoing Einstein QNM multiplied by time, even though the constant generalized component has weakened asymptotic falloff.

This gives a Schwarzschild counterpart, in asymptotically flat spacetime, of the critical-gravity logarithmic-partner mechanism. The open step is not the existence of the defective radial response; it is the construction of a global causal domain that accommodates the differentiated endpoint state and supports a justified retarded contour deformation.

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